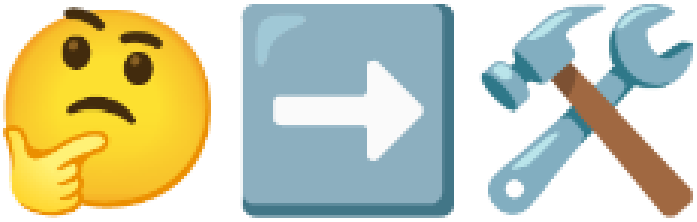
🛠️ What Problem Can We Solve? 🛠️

Engineering Design — Unit 1, Lesson 1 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–40 minutes 🧪 **Subject:** Engineering Design



Engineers start by noticing a problem and imagining a way to help.

📋 **NGSS Standards:**
K-2-ETS1-1 — Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.

🎯 Learning Objective

Students will **identify** a simple everyday problem and **describe** who needs help and what a successful solution should do.

💡 The Big Idea

Engineering begins with a problem. A problem is something that is hard to do or something we want to make better. Engineers look carefully, ask questions, and think about how to help people.

> 💡 **Key vocabulary:** problem, solution, engineer, improve, tool, help

📦 Materials

- Pencil
- Crayons
- Worksheet pages from this lesson
- Optional: classroom or household objects to discuss

🔬 Let's Get Started

Step 1 — Notice Problems (5 min)
Talk about everyday examples: carrying many books, keeping papers from blowing away, reaching something high, keeping lunch cold. Ask your child which ones sound like real problems.

Step 2 — Who Needs Help? (8 min)
Pick one simple problem together. Ask: "Who has this problem?" and "What is hard about it?" Help the child explain the situation in plain words.

Step 3 — What Would Help? (8 min)
Ask: "What should a helpful tool or object do?" Focus on the job of the solution, not the fancy details. For example: *"It should help carry the books without dropping them."*

Step 4 — Draw the Problem (8 min)
Have the child draw the problem and circle the part that needs to change or improve.

💬 Discussion Questions

🗣️ **Question 1:** "What makes something a problem?"

💬 *A problem is something that is hard to do, something unsafe, or something we want to make better.*

🗣️ **Question 2:** "Who are we trying to help?"

💬 *Engineers help people. It is important to know who the user is.*

🗣️ **Question 3:** "What should a good solution do?"

💬 *It should help with the problem. It does not need to be perfect at first, but it should do its job. This is the design* ***success rule*** *—what the tool must do to be helpful.*

👩‍🏫 Parent/Teacher Notes

- Keep problems simple and familiar.
- Do not rush into building. This lesson is about defining the problem clearly.
- Strong student response: "The problem is ___ because ___. A solution should help by ___."

🛠️ Student Worksheet — What Problem Can We Solve?

Name: _____ Date: _____

1. What is one problem you want to solve?

2. Who has this problem?

3. What should a good solution do?

4. Draw the problem:

Draw here ✏️

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers will vary. Accept any realistic everyday problem with a clear person/user and a sensible description of what a solution should do.

Sample response: *"The problem is carrying too many books. The user is a student. A helpful solution should hold the books together and stop them from falling."*

Little Lab Coats

🦆 Look at Animal Solutions 🦔

Engineering Design — Unit 1, Lesson 2 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–40 minutes 🧪 **Subject:** Engineering Design

Nature is full of structures that solve problems.

📋 **NGSS Standards:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **observe** animal parts that solve problems and **describe** how those parts could inspire human-made designs to solve a human problem.

💡 The Big Idea

Animals have body parts that help them solve problems. Ducks have webbed feet for swimming. Hedgehogs have spines for protection. Birds have beaks that help them get food. Engineers can look at these structures and borrow ideas from nature.

💡 **Key vocabulary:** mimic, structure, protect, move, carry, solve

📦 Materials

- Pencil and crayons
- Worksheet pages

- Optional pictures of ducks, birds, turtles, hedgehogs, burrs, or seed pods

🔬 Let's Get Started

Step 1 — Notice Animal Parts (8 min)
Look at a few animal examples together. Ask: "What body part do you notice?" and "What job does it do?"

Step 2 — Match Structure to Job (8 min)
Talk about a few examples:

- Webbed feet help animals swim.
- Claws help animals dig or climb.
- Beaks help birds pick up food.
- Spines help animals protect themselves.

Step 3 — Borrow the Idea (8 min)
Ask: "How could a person use an idea like that?" Example: webbed hands for swimming gloves, a claw-like grabber tool, a protective helmet inspired by shells. Remind your child they do not need a perfect or realistic invention—just a clear nature idea that could help a person.

Step 4 — Draw a Nature-Inspired Idea (8 min)
Have your child draw one animal part and one tool or object inspired by it.

💬 Discussion Questions

🗣️ **Question 1:** "What problem does this animal part help solve?"

💬 *It might help with moving, getting food, staying safe, or carrying seeds.*

🗣️ **Question 2:** "How could a human tool copy that idea?"

💬 *A good answer explains what job the tool would do, not just what it looks like.*

🗣️ **Question 3:** "Why do engineers look at nature?"

💬 *Nature already has many useful solutions. Engineers can learn from them.*

👩‍🏫 **Parent/Teacher Notes:**

- This lesson bridges life science and engineering.
- Children do not need advanced biomimicry language; "copying an idea from nature" is enough.

- Strong student outcome: naming one animal structure and one matching human solution idea.

🦆 Student Worksheet — Look at Animal Solutions

Name: _____ Date: _____

1. What animal part did you notice?

2. What job does it do?

3. What tool could copy that idea?

4. Draw the animal and your new tool:

Draw here ✏️

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Accept any sensible pairing of animal structure + job + human design idea.

Sample pairing: *Duck feet help with swimming → swimming gloves that help a person move through water.*

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✏️ Plan a Nature-Inspired Tool ✏️

Engineering Design — Unit 1, Lesson 3 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–40 minutes 🧪 **Subject:** Engineering Design

📋 **NGSS Standards:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **plan** a simple tool inspired by an animal or plant part, **label** how the design will help solve a problem, and name what the tool must do to be helpful.

💡 The Big Idea

Before engineers build, they make a plan. A plan helps us think about what the tool should look like, what materials we might use, and what job it should do. Good plans are simple and clear.

💡 **Key vocabulary:** plan, label, design, build, mimic, idea

🎯 **Success rule:** A good plan should show what the tool must do—for example, grab, hold, protect, or scoop.

📦 Materials

- Pencil
- Crayons
- Worksheet pages
- Optional sample building materials for discussion

🔬 Let's Get Started

Step 1 — Review Yesterday's Idea (5 min)
Talk about the animal part or plant part the child liked best yesterday.

Step 2 — Plan the Tool (10 min)
Ask: "What should your tool do?" and "What shape or part from nature are you copying?" Sentence frame: *"My tool will help ___ by using a shape like ___."*

Step 3 — Draw and Label (12 min)
Have the child draw the tool and label the important part. Encourage simple words like grab, protect, scoop, slide, hold.

Step 4 — Explain the Plan (8 min)
Have the child explain how the tool will help someone and what part came from nature.

💬 Discussion Questions

🗣️ **Question 1:** "What job should your tool do?"

💬 *A strong answer is about the job, like holding, scooping, protecting, or helping someone move something.*

🗣️ **Question 2:** "What natural part gave you the idea?"

💬 *Students should connect the design to webbed feet, claws, shells, burrs, stems, or another structure.*

🗣️ **Question 3:** "Why is it helpful to make a plan before building?"

💬 *A plan helps us think first and gives us something to follow when we build.*

👩‍🏫 **Parent/Teacher Notes:**

- This lesson should feel calm and thoughtful.
- Simple labeled drawings are enough.
- What matters most is the connection between the problem, the natural structure, and the planned solution.

✏️ Student Worksheet — Plan a Nature-Inspired Tool

Name: _____ Date: _____

1. What problem will your tool solve?

2. What plant or animal part gave you the idea?

3. Draw and label your tool:

Draw your plan here ✏️

4. My tool will help by:

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Accept any clear plan that links a natural structure to a simple tool idea and explains the job the tool should do.

Sample plan: *"My tool will help pick up blocks by using a bird-beak shape. The front part closes to grab the block."*

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🔨 Build Your First Model 🔨

Engineering Design — Unit 1, Lesson 4 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 35–45 minutes 🧪 **Subject:** Engineering Design

📋 **NGSS Standards:**
K-2-ETS1-2 — Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.

🎯 Learning Objective

Students will **build** a first model of their nature-inspired design, **compare** it to their original plan, and explain how it is supposed to help the user.

💡 The Big Idea

A model is a first version. Engineers build models so they can learn what works and what needs to change. A first model does not need to be perfect, but it should still try to do the important job from the plan.

💡 **Key vocabulary:** model, build, test, compare, shape, function

🎯 **Design reminder:** Before building, ask: *Who is this for, and what must it help them do?*

📦 Materials

- Paper
- Tape
- Scissors
- Craft sticks or cardboard scraps
- Pipe cleaners, string, or other simple classroom materials
- Worksheet pages

🔬 Let's Get Started

Step 1 — Review the Plan (5 min)
Look at the drawing from Lesson 3. Ask: "What is the most important part to build first?"

Step 2 — Build the Model (15 min)
Use simple materials to build the first version. Remind students that it is okay if it does not look exactly like the drawing.

Step 3 — Compare Plan and Model (8 min)
Ask: "What stayed the same? What changed? Which change helped your model do its job?"

Step 4 — Get Ready to Test (5 min)
Talk about how they will know if the model helps solve the problem.

💬 Discussion Questions

🗣️ **Question 1:** "What part of your model is most important?"

💬 *Students should identify the part that does the main job.*

🗣️ **Question 2:** "How is your model the same as your drawing?"

💬 *They can talk about shape, parts, or the job it is supposed to do.*

🗣️ **Question 3:** "Why is a first model allowed to be imperfect?"

💬 *Because engineers learn by building and then improving.*

👩‍🏫 **Parent/Teacher Notes:**

- Do not over-help. Let the child own the design.
- Messy but thoughtful is fine.
- Today is about turning a plan into something real.

🔨 Student Worksheet — Build Your First Model

Name: _____ Date: _____

1. Draw your model after building it:

Draw here ✏️

2. What stayed the same from your plan?

3. What changed?

4. How will you test it?

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A strong response shows a real connection between the planned design, the built model, and an idea for testing. Example: *"My drawing had a wide scoop to help pick up blocks. My model still has the scoop, but I made the handle shorter so it is easier to hold. I will test whether it can pick up the block without dropping it."*

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🧪 Test and Improve Your Design 🧪

Engineering Design — Unit 1, Lesson 5 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–40 minutes 🧪 **Subject:** Engineering Design

📋 **NGSS Standards:**
K-2-ETS1-3 — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

🎯 Learning Objective

Students will **test** a model, **compare** how version 1 and version 2 perform, and **make one improvement** based on what they noticed.

💡 The Big Idea

Engineers test their ideas. A test gives us information. If a design does not work the first time, that is not failure — it is feedback that helps us improve.

💡 **Key vocabulary:** test, improve, strength, weakness, compare, change

📦 Materials

- The model from Lesson 4
- Same building materials used before
- Worksheet pages

🔬 Let's Get Started

Step 1 — Decide on a Fair Test (5 min)
Talk about the job of the design and how to check whether it can do that job.

Step 2 — Test the Model (10 min)
Try the design. Watch carefully. Did it help solve the problem? What part worked well? What part needs help?

Step 3 — Change One Thing (8 min)
Choose one improvement. Keep the rest the same so you can tell whether the change helped.

Step 4 — Test Again (8 min)
Run the test again and compare the new result to the first one. Ask: *"Did version 2 do the job better, worse, or about the same? What evidence do you have?"*

💬 Discussion Questions

🗣️ **Question 1:** "What worked well in your first model?"
💬 *A strength is something the design already does well.*

🗣️ **Question 2:** "What needed to change?"
💬 *A weakness is something that did not help enough or made the tool harder to use.*

🗣️ **Question 3:** "Did your change make the design better? How do you know?"
💬 *The best answers use evidence from the test.*

👩‍🏫 **Parent/Teacher Notes:**

- Keep tests simple and visible.
- Students only need one meaningful improvement.
- We are teaching the cycle: build → test → improve.

🧪 Student Worksheet — Test and Improve Your Design

Name: _____ Date: _____

Version	What happened?	What changed?	Did it work better?
Version 1			
Version 2			

What was one strength of your design?

What was one weakness?

How do you know version 2 worked better or worse?

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A strong worksheet shows an actual test, one change, and a comparison between before and after.

- **Sample strength:** “The tool could pick up the object.”
- **Sample weakness:** “The handle bent and made it hard to use.”
- **Sample comparison:** *"Version 2 worked better than version 1 because I made the handle shorter, and it held the object without dropping it."*

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🎤 Share Your Final Design 🎤

Engineering Design — Unit 1, Lesson 6 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–40 minutes 🧪 **Subject:** Engineering Design

📋 **NGSS Standards:**
K-2-ETS1-3 — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

🎯 Learning Objective

Students will **share** their final design, **explain** how it solves the problem, and **compare** how the final version works better than the first version.

💡 The Big Idea

Engineers share their ideas with other people. Explaining the problem, the design, and the improvement helps others understand how the solution works.

💡 **Key vocabulary:** final design, explain, improve, share, solve, evidence

📦 Materials

- Final design model
- Worksheet pages
- Optional: family member or partner audience

🔬 Let's Get Started

Step 1 — Get Ready to Present (8 min)
Review the original problem, the first idea, and the change that made the design better.

Step 2 — Share the Final Design (10 min)
Have the child show the model and explain what it does, what natural idea inspired it, and what changed after testing.

Step 3 — Ask Questions (8 min)
Listeners can ask: "What problem does it solve?" "What part is most important?" "What did you improve?"

Step 4 — Celebrate the Process (5 min)
Emphasize that the goal was not perfection. The goal was to think like an engineer.

💬 Discussion Questions

🗣️ **Question 1:** "What problem does your design solve?"

💬 *The problem should be clear and specific.*

🗣️ **Question 2:** "What part of nature inspired your design?"

💬 *Students should name the plant or animal structure they copied.*

🗣️ **Question 3:** "How did your design get better from the first version to the final version?"

💬 *A strong answer names at least one improvement and what effect it had. Try the sentence frame: "**I changed ___, and now my design ___.**"*

👩‍🏫 **Parent/Teacher Notes:**

- This lesson is about communication and reflection.
- Take photos if you want a record of the child's engineering work.
- The best presentations explain the process, not just the final object.

🎤 Student Worksheet — Share Your Final Design

Name: _____ Date: _____

1. My problem was:

2. My design was inspired by:

3. One improvement I made was:

4. I changed _____, and now my design _____.

5. Draw your final design:

Draw here ✏️

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A complete response identifies the problem, the inspiration from nature, and at least one meaningful improvement.

Sample complete response: *"My problem was picking up blocks. My design was inspired by a bird beak. I made the grabbing part narrower, and now my design can hold the block better than my first version."*